Sales Forecasting & Demand Intelligence System

Executive Business Report

This report summarizes the findings of an internal analytics project that examined four years of retail sales data to understand demand patterns, forecast future sales, flag unusual sales behavior, and group products by how predictable their demand is. The goal is to help supply chain and inventory decisions be based on data rather than guesswork.

Key Findings from Historical Data

- Technology is the top revenue-generating category, followed by Furniture, with Office Supplies contributing the least.
- The West region generates the highest total sales, followed by East; the South region trails all others.
- East has the most consistent year-over-year growth, while West grows faster on average but with much larger swings — meaning West's growth is real but harder to plan around precisely.
- Sales are strongly seasonal: November and December consistently spike every year, driven by holiday shopping demand.
- Shipping times are fairly uniform across regions (roughly 3.9 to 4.1 days on average), so delivery speed is not currently a regional pain point.

3-Month Sales Forecast

Three forecasting approaches were built and tested — SARIMA (a statistical model), Prophet (an industry-standard forecasting tool), and XGBoost (a machine-learning model). Each was scored on how closely its predictions matched actual historical sales that were deliberately held back for testing.

Model	MAE	RMSE	MAPE	Recommended
SARIMA	**14,860.76**	**17,719.90**	**26.33%**	**✓ Selected**
Prophet	17,360.04	21,571.07	28.91%	
XGBoost	23,598.36	26,657.31	34.68%	

SARIMA produced the lowest forecasting error across all three metrics and is the recommended model for production use. Applied at the category and region level, the strongest projected near-term growth is in Technology and Office Supplies, while Furniture is expected to remain comparatively flat over the next quarter.

Top Anomalies Detected

Two independent anomaly detection methods (Isolation Forest and Z-Score) were applied to weekly sales. Five weeks were flagged by both methods, giving high confidence they represent genuine, unusual sales events rather than statistical noise.

- Late-2018 spikes (early November and early December) align with expected holiday demand and should be treated as a recurring, planned event each year.
- A March 2015 spike does not match the usual seasonal pattern and likely reflects a one-off cause — such as a bulk order or short-term promotion — that would benefit from a manual follow-up with the sales team.

- A supplementary check on an unrelated dataset (industry-wide video game sales) confirmed the same anomaly detection approach reliably identifies genuine turning points, not just artifacts of the Superstore data specifically.

Product Demand Segmentation

Products were grouped into four demand behavior clusters using sales volume, growth, volatility, and average order value, to guide stocking strategy:

- **High Volume, Stable Demand (Accessories, Binders, Chairs, Phones, Storage, Tables): safe to stock on standard replenishment cycles.**
- **Low Volume, Stable Demand (8 sub-categories incl. Art, Paper, Envelopes): keep lean inventory; stockouts here are low-cost.**
- **High Value, High Volatility (Copiers, Machines): keep safety stock buffers; large individual orders make demand swing unpredictably.**
- **Emerging/Growing Niche (Supplies): monitor before committing significant inventory — its high growth rate is partly a small-base effect.**

Business Recommendations

1. Prioritize Technology and West-region inventory planning around the SARIMA forecast, and build in extra safety stock for the Nov–Dec holiday surge every year.
2. Apply a demand-triggered (rather than fixed-cycle) reorder strategy for Copiers and Machines, since their high order value combined with high volatility makes fixed forecasts unreliable for this group.
3. Investigate the March 2015 sales spike directly with the sales team to determine its cause; if it was a repeatable promotional tactic, consider running it deliberately in future off-peak periods.

Risk & Limitation

Forecasts built separately for smaller regional segments (e.g. individual regions) are less stable than the overall company-wide forecast — in one case, the model even predicted an unrealistic negative sales figure for a low-volume period. Segment-level forecasts should be treated as directional guidance rather than precise targets, particularly for smaller or more volatile product and region combinations, until more historical data is available.